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Fuel for Exercise: Bioenergetics and Muscle Metabolism

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In Closing

“H

itting the wall” is a common expression heard among marathon runners, and more than half of all nonelite marathon runners report having hit the wall during a marathon regardless of how hard they trained. This phenomenon usually happens around mile 20 to 22. The runner’s pace slows considerably and the legs feel like lead. Tingling and numbness are often felt in the legs and arms, and thinking often becomes fuzzy and confused. Hitting the wall is basically running out of available energy.

The runner’s primary fuel sources during prolonged exercise are carbohydrates and fats. Fats might seem to be the logical first choice of fuel for endurance events—they are ideally designed to be energy dense, and stores are virtually unlimited. Unfortunately, fat metabolism requires a constant supply of oxygen, and delivery of energy is slower than that provided by carbohydrate metabolism.

Most runners are able to store 2,000 to 2,200 calories of glycogen in their liver and muscles, which is enough to provide energy for about 20 mi (32 km) of moderate-pace running. Since the body is much less efficient at converting fat to energy, running pace slows and the runner suffers from fatigue. Furthermore, carbohydrates are the sole fuel source for brain function. Physiology, not coincidence, dictates why so many marathon runners hit the wall at around the 20 mi mark.

Chemical reactions in plants (photosynthesis) convert light from the sun into stored chemical energy. In turn, humans obtain energy by eating either plants or animals that feed on plants. Nutrients from ingested foods are provided in the form of carbohydrates, fats, and proteins. These three basic fuels, or energy **substrates**, can ultimately be broken down to release the stored energy. Each cell contains chemical pathways that convert these substrates to energy that can then be used by that cell and other cells of the body, a process called **bioenergetics**. All of the chemical reactions in the body are collectively called **metabolism**.

Because all energy eventually degrades to heat, the amount of energy released in a biological reaction can be measured from the amount of heat produced. Energy in biological systems is measured in calories. By definition, 1 calorie (cal) equals the amount of heat energy needed to raise 1 g of water 1 °C, from 14.5 °C to 15.5 °C. In humans, energy is expressed in **kilocalories (kcal)**, where 1 kcal is

the equivalent of 1,000 cal. Sometimes the term *Calorie* (with a capital C) is used synonymously with kilocalorie, but *kilocalorie* is more technically and scientifically correct. Thus, when one reads that someone eats or expends 3,000 Cal per day, it really means the person is ingesting or expending 3,000 *kcal* per day.

Some free energy in the cells is used for growth and repair throughout the body. Such processes build muscle mass during training and repair muscle damage after exercise or injury. Energy also is needed for active transport of many substances, such as sodium, potassium, and calcium ions, across cell membranes to maintain homeostasis. Myofibrils use energy to cause sliding of the actin and myosin filaments, resulting in muscle action and force generation, as described in [chapter 1](#).

Energy Substrates

Energy is released when chemical bonds—the bonds that hold elements together to form molecules—are broken. Substrates are composed primarily of carbon, hydrogen, oxygen, and (in the case of protein) nitrogen. The molecular bonds that hold these elements together are relatively weak and therefore provide little energy when broken. Consequently, food is not used directly for cellular operations. Rather, the energy in food's molecular bonds is chemically released within our cells and then stored in the form of the high-energy compound introduced in [chapter 1](#), adenosine triphosphate (ATP), which is discussed in detail later in this chapter.

At rest, the energy that the body needs is derived almost equally from the breakdown of carbohydrates and fats. Proteins serve important functions as enzymes that aid chemical reactions and as structural building blocks but usually provide little energy for metabolism. During intense, short-duration muscular effort, more carbohydrate is used, with less reliance on fat to generate ATP. Longer, less intense exercise uses both carbohydrate and fat for sustained energy production.

Carbohydrate

The amount of **carbohydrate** used during exercise is related to both the carbohydrate availability and the muscles' well-developed system for carbohydrate metabolism. All carbohydrates are ultimately

converted to the simple six-carbon sugar, **glucose** (figure 2.1), a monosaccharide (one-unit sugar) that is transported through the blood to all body tissues. Under resting conditions, ingested carbohydrate is stored in muscles and liver in the form of a more complex polysaccharide (multiple linked sugar molecules), **glycogen**. Glycogen is stored in the cytoplasm of muscle cells until those cells use it to form ATP. Additional glycogen stored in the liver is converted back to glucose as needed and then transported by the blood to active tissues, where it is metabolized.

Muscle and liver glycogen stores are limited, especially if the diet contains an insufficient amount of carbohydrate, and can be depleted during prolonged, intense exercise. Thus, we rely heavily on dietary sources of starches and sugars to continually replenish our carbohydrate reserves. Without adequate carbohydrate intake, muscles can be deprived of their primary energy source. Furthermore, carbohydrates are the only energy source used by brain tissue; therefore, severe carbohydrate depletion results in negative cognitive effects.

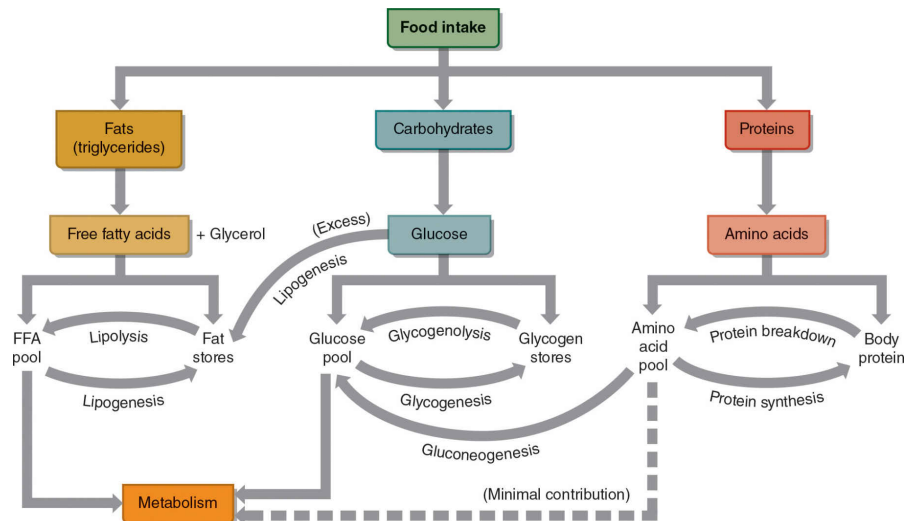


FIGURE 2.1 Cellular metabolism results from the breakdown of three fuel substrates provided by the diet. Once each is converted to its usable form, it either circulates in the blood as an available “pool” to be used for metabolism or is stored in the body.



Fat

Fats provide a large portion of the energy used during prolonged, less intense exercise. Body stores of potential energy in the form of fat are substantially larger than the reserves of carbohydrate, in terms of both weight and energy availability. [Table 2.1](#) provides an indication of the total body stores of these two energy sources in a lean person (12% body fat). For the average middle-aged adult with more body fat (adipose tissue), the fat stores would be approximately twice as large, whereas the carbohydrate stores would be about the same. But fat is less readily available for cellular metabolism because it must first be reduced from its complex form, **triglyceride**, to its basic components, glycerol and **free fatty acids (FFAs)**. Only FFAs are used to form ATP ([figure 2.1](#)).

TABLE 2.1 Body Stores of Fuels and Associated Energy Availability

Location	g	kcal
Carbohydrate		
Liver glycogen	110	451
Muscle glycogen	500	2,050
Glucose in body fluids	15	62
Fat		
Subcutaneous and visceral	7,800	73,320
Intramuscular	161	1,513
Total	7,961	74,833

Note. These estimates are based on a body weight of 65 kg (143 lb) with 12% body fat.

Substantially more energy is derived from breaking down a gram of fat (9.4 kcal/g) than from the same amount of carbohydrate (4.1 kcal/g). Nonetheless, the rate of energy release from fat is too slow to meet all of the energy demands of intense muscular activity.

Other types of fats found in the body serve non-energy-producing functions. Phospholipids are a key structural component of all cell membranes and form protective sheaths around some large nerves. Steroids are found in cell membranes and also function as hormones or as building blocks of hormones, such as estrogen and testosterone.

Protein

Protein can be used as a minor energy source under some circumstances, but it must first be converted into glucose (figure 2.1). In the case of severe energy depletion or starvation, protein may even be used to generate FFAs for energy. The process by which protein or fat is converted into glucose is called **gluconeogenesis**. The process of converting protein into fatty acids is termed **lipogenesis**. Protein can supply up to 10% of the energy needed to sustain prolonged exercise. Only the most basic units of protein—the amino acids—can be used for energy. A gram of protein yields about 4.1 kcal.

Controlling the Rate of Energy Production

To be useful, free energy must be released from chemical compounds at a controlled rate. This rate is determined primarily by two things, the availability of the primary substrate and enzyme activity. The availability of large amounts of a substrate increases the activity of that particular pathway. An abundance of one particular fuel (e.g., carbohydrate) can cause cells to rely more on that source than on alternatives. This influence of substrate availability on the rate of metabolism is termed the *mass action effect*.

Specific protein molecules called **enzymes** also control the rate of free-energy release. Many of these enzymes speed up the breakdown (**catabolism**) of chemical compounds. Chemical reactions occur only when the reacting molecules have sufficient initial energy to start the reaction or chain of reactions. Enzymes do not cause a chemical reaction to occur and do not determine the amount of usable energy that is produced by these reactions. Rather, they speed up reactions by lowering the **activation energy** that is required to begin the reaction (figure 2.2).

Although the enzyme names are quite complex, most end with the suffix *-ase*. For example, an important enzyme that breaks down ATP and releases stored energy is adenosine triphosphatase, better known as ATPase.

Biochemical pathways that result in the production of a product from a substrate almost always involve multiple steps. Each individual step is typically catalyzed by a specific enzyme. Therefore, increasing

the amount of enzyme present or the activity of that enzyme (for example, by changing the temperature or pH) results in an increased rate of product formation through that metabolic pathway. Additionally, many enzymes require other molecules called cofactors to function, so cofactor availability may also affect enzyme function and therefore the rate of metabolic reactions.

As illustrated in [figure 2.3](#), metabolic pathways typically have one enzyme that is of particular importance in controlling the reaction's overall rate. This enzyme, usually located in an early step in the pathway, is known as the **rate-limiting enzyme**. The activity of a rate-limiting enzyme is determined by the accumulation of substances farther down the pathway that decrease enzyme activity through **negative feedback**.

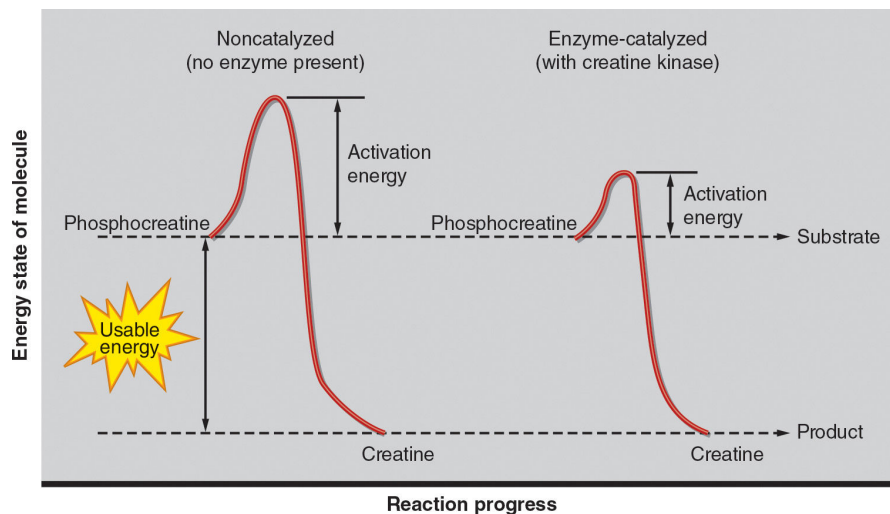


FIGURE 2.2 Enzymes control the rate of chemical reactions by lowering the activation energy required to initiate the reaction. In this example, the enzyme creatine kinase binds to its substrate phosphocreatine to increase the rate of production of creatine.

Adapted from original figure provided by Dr. Martin Gibala, McMaster University, Hamilton, Ontario, Canada.



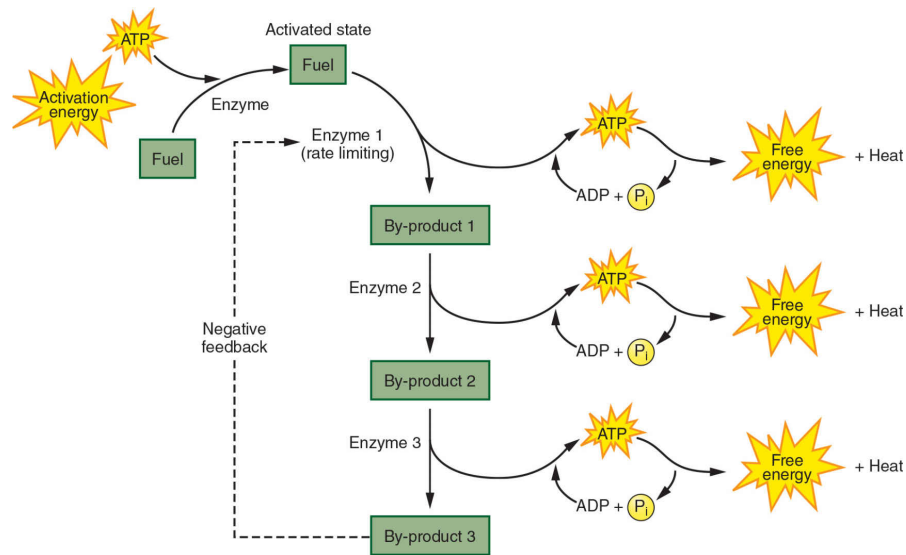


FIGURE 2.3 A typical metabolic pathway showing the important role of enzymes in controlling the rate of the reaction. An input of energy in the form of stored adenosine triphosphate (ATP) is needed to begin the series of reactions (activation energy), but less initial energy is needed if one or more enzymes are involved in this activation step. As fuels are subsequently degraded into by-products along the metabolic pathway, ATP is formed. Utilization of the stored ATP results in the release of usable energy, heat, and the release of adenosine diphosphate (ADP) and inorganic phosphate (P_i).



In Review

- Energy for cell metabolism is derived from three substrates in foods: carbohydrate, fat, and protein. Proteins provide little of the energy used for metabolism under normal conditions.
- Within cells, the usable storage form of the energy we derive from food is the high-energy compound adenosine triphosphate, or ATP.
- Carbohydrate and protein each provide about 4.1 kcal energy per gram, compared with about 9.4 kcal/g for fat.
- Carbohydrate, stored as glycogen in muscle and the liver, is more quickly accessible as an energy source than either protein or fat. Glucose, directly from

food or broken down from stored glycogen, is the usable form of carbohydrate.

- Fat, stored as triglycerides in adipose tissue, is an ideal storage form of energy. Free fatty acids from the breakdown of triglycerides are converted to energy.
 - Carbohydrate stores in the liver and skeletal muscle are limited to about 2,500 to 2,600 kcal of energy, or the equivalent of the energy needed for about 40 km (25 mi) of running. Fat stores can provide more than 70,000 kcal of energy.
 - Enzymes control the rate of metabolism and energy production. Enzymes can speed up the overall reaction by lowering the initial activation energy and by catalyzing various steps along the pathway.
 - Enzymes can be inhibited through negative feedback of subsequent pathway by-products (or often ATP), slowing the overall rate of the reaction. This usually involves a particular enzyme located early in the pathway called the rate-limiting enzyme.
-

One example of a substance that may accumulate and feed back to decrease enzyme activity would be the end product of the pathway; another would be ATP and its breakdown products, ADP and inorganic phosphate. If the goals of a metabolic pathway are to form a chemical product and release free energy in the form of ATP, it makes sense that an abundance of either that end product or ATP would feed back to slow further production and release, respectively.

VIDEO 2.1 Presents Mark Hargreaves discussing the sensitivity of ATP production to muscle activity and control of ATP production during exercise.



Storing Energy: High-Energy Phosphates

The immediately available source of energy for almost all bodily functions, including muscle contraction, is ATP. An ATP molecule ([figure 2.4a](#)) is composed of adenosine (a molecule of adenine joined to a molecule of ribose) combined with three inorganic phosphate (P_i) groups. Adenine is a nitrogen-containing base, and ribose is a five-carbon sugar. When an ATP molecule is combined with water (hydrolysis) and acted on by the enzyme ATPase, the last phosphate group splits away, rapidly releasing a large amount of free energy (approximately 7.3 kcal per mole of ATP under standard conditions, but possibly up to 10 kcal per mole of ATP or greater within the cell). This reduces the ATP to **adenosine diphosphate (ADP)** and P_i ([figure 2.4b](#)).

To generate ATP, a phosphate group is added to the relatively low-energy compound, ADP, in a process called **phosphorylation**. This process requires a considerable amount of energy. Some ATP is generated independent of oxygen availability, and such metabolism is called substrate-level phosphorylation. Other ATP-producing reactions (discussed later in the chapter) occur without oxygen, while still others occur with the aid of oxygen, a process called **oxidative phosphorylation**.

As shown in [figure 2.3](#), ATP is formed from ADP and P_i via phosphorylation as fuels are broken down into fuel by-products at various steps along a metabolic pathway. The storage form of energy, ATP, can subsequently release free or usable energy when needed as it is once again broken down into ADP and P_i .

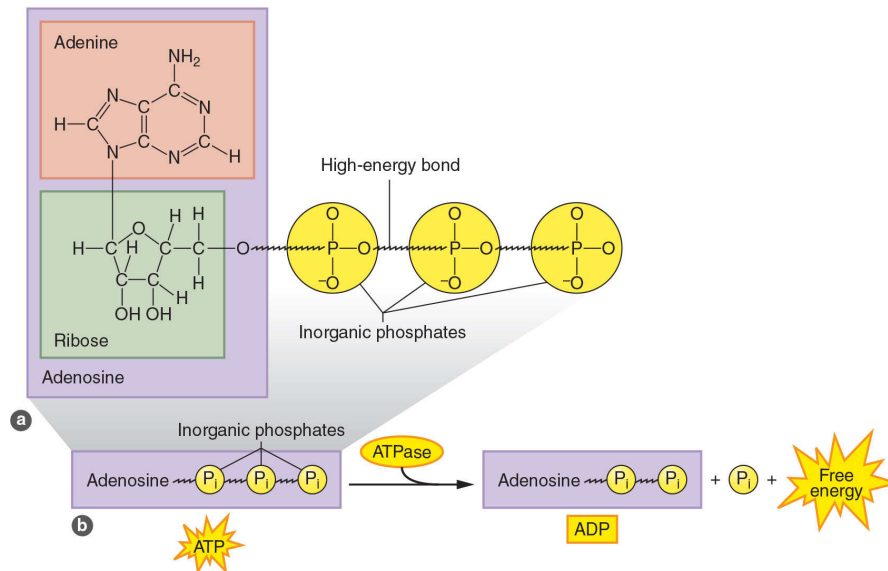


FIGURE 2.4 (a) The structure of an adenosine triphosphate (ATP) molecule, showing the high-energy phosphate bonds. (b) When the third phosphate on the ATP molecule is separated from adenosine by the action of adenosine triphosphatase (ATPase), energy is released.



The Basic Energy Systems

Cells can store only very limited amounts of ATP and must constantly generate new ATP to provide needed energy for all cellular metabolism, including muscle contraction. Cells generate ATP through any one of (or a combination of) three metabolic pathways:

1. The ATP-PCr system
2. The glycolytic system (glycolysis)
3. The oxidative system (oxidative phosphorylation)

The first two systems can act in the absence of oxygen and are jointly termed **anaerobic metabolism**. The third system requires oxygen and therefore comprises **aerobic metabolism**.

ATP-PCr System

The simplest of the energy systems is the **ATP-PCr system**, shown in [figure 2.5](#). In addition to storing a very small amount of adenosine

triphosphate (ATP) directly, cells contain another high-energy phosphate molecule that stores energy called **phosphocreatine (PCr)**; sometimes called creatine phosphate). This simple pathway involves donation of a P_i from PCr to ADP to form ATP. Unlike what occurs with the limited freely available ATP in the cell, energy released by the breakdown of PCr is not directly used for cellular work. Instead, it regenerates ATP to maintain a relatively constant supply under resting conditions.

The release of energy from PCr is catalyzed by the enzyme **creatine kinase**, which acts on PCr to separate P_i from creatine. The energy released can then be used to add a P_i molecule to an ADP molecule, forming ATP. As energy is released from ATP by the splitting of a phosphate group, cells can prevent ATP depletion by breaking down PCr, providing energy and P_i to re-form ATP from ADP.

Following the principles of negative feedback and rate-limiting enzymes discussed earlier, creatine kinase activity is enhanced when concentrations of ADP or P_i increase, and is inhibited when ATP concentrations increase. When intense exercise is initiated, the small amount of available ATP in muscle cells is broken down for immediate energy, yielding ADP and P_i . The increased ADP concentration enhances creatine kinase activity, and PCr is catabolized to form additional ATP. As exercise progresses and additional ATP is generated by the other two energy systems—the glycolytic and oxidative systems—creatine kinase activity is inhibited.

This process of breaking down PCr to allow formation of ATP is rapid and can be accomplished without any special structures within the cell. The ATP-PCr system is classified as substrate-level metabolism. Although it can act in the presence of oxygen, the process does not require oxygen.

During the first few seconds of intense muscular activity, such as sprinting, ATP is maintained at a relatively constant level, but PCr declines steadily as it is used to replenish the depleted ATP (see [figure 2.6](#)). At exhaustion, however, both ATP and PCr levels are low and are unable to provide energy for further muscle contraction and relaxation. Thus, the capacity to maintain ATP levels with the energy from PCr is limited. The combination of ATP and PCr stores can

sustain the muscles' energy needs for only 3 to 15 s during an all-out sprint. Beyond that time, muscles must rely on other processes for ATP formation: glycolytic and oxidative pathways.

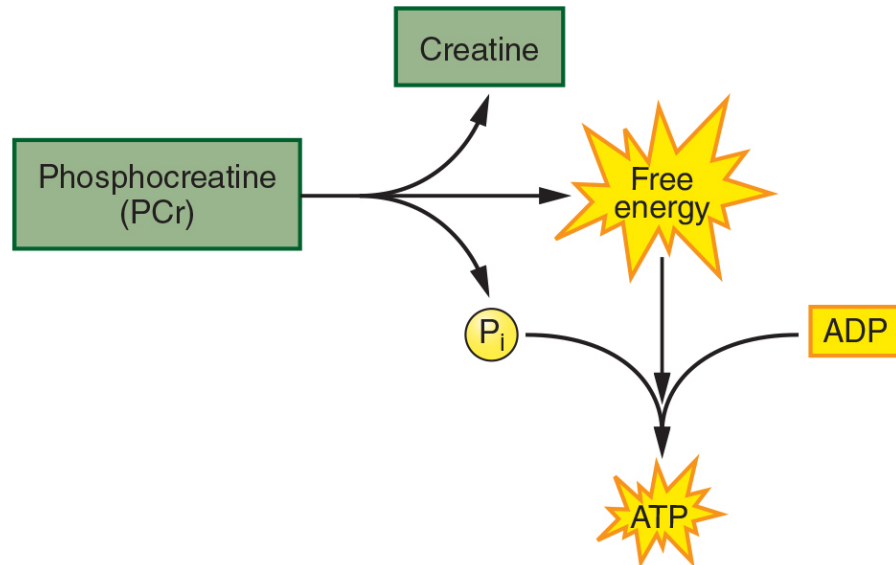


FIGURE 2.5 In the ATP-PCr system, adenosine triphosphate (ATP) can be recreated via the binding of an inorganic phosphate (P_i) to adenosine diphosphate (ADP) with the energy derived from the breakdown of phosphocreatine (PCr).



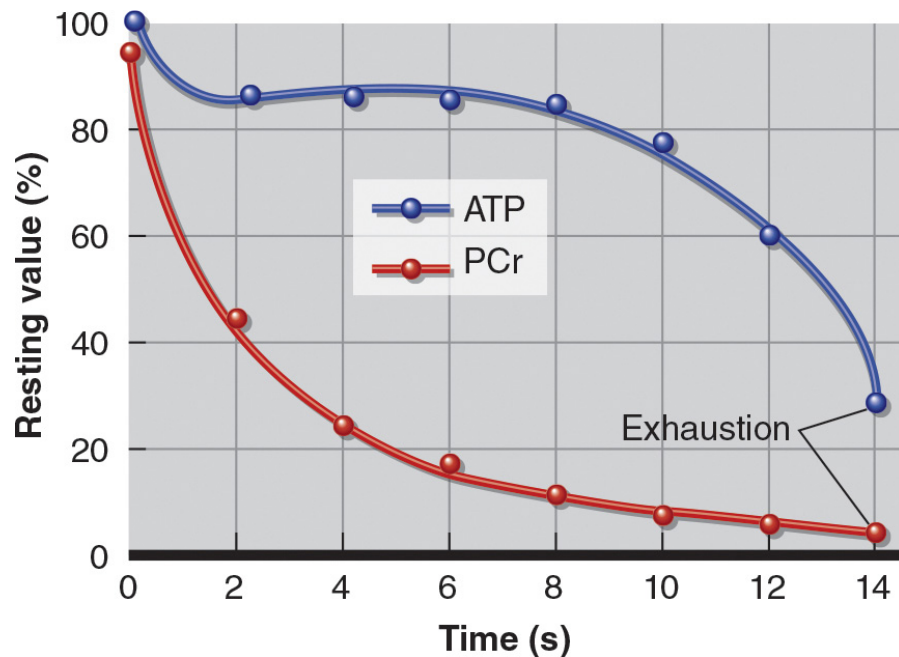


FIGURE 2.6 Changes in type II (fast-twitch) skeletal muscle adenosine triphosphate (ATP) and phosphocreatine (PCr) during 14 s of maximal muscular effort (sprinting). Although ATP is being used at a very high rate, the energy from PCr is used to synthesize ATP, initially preventing the ATP level from decreasing. However, at exhaustion, both ATP and PCr levels are low.



Glycolytic System

The ATP-PCr system has a limited capacity to generate ATP for energy, lasting only a few seconds. The second method of ATP production involves the liberation of energy through the breakdown (“lysis”) of glucose. This system is called the glycolytic system because it entails **glycolysis**, the breakdown of glucose through a pathway that involves a sequence of glycolytic enzymes. Glycolysis is a more complex pathway than the ATP-PCr system, and the sequence of steps involved in this process is presented in [figure 2.7](#).

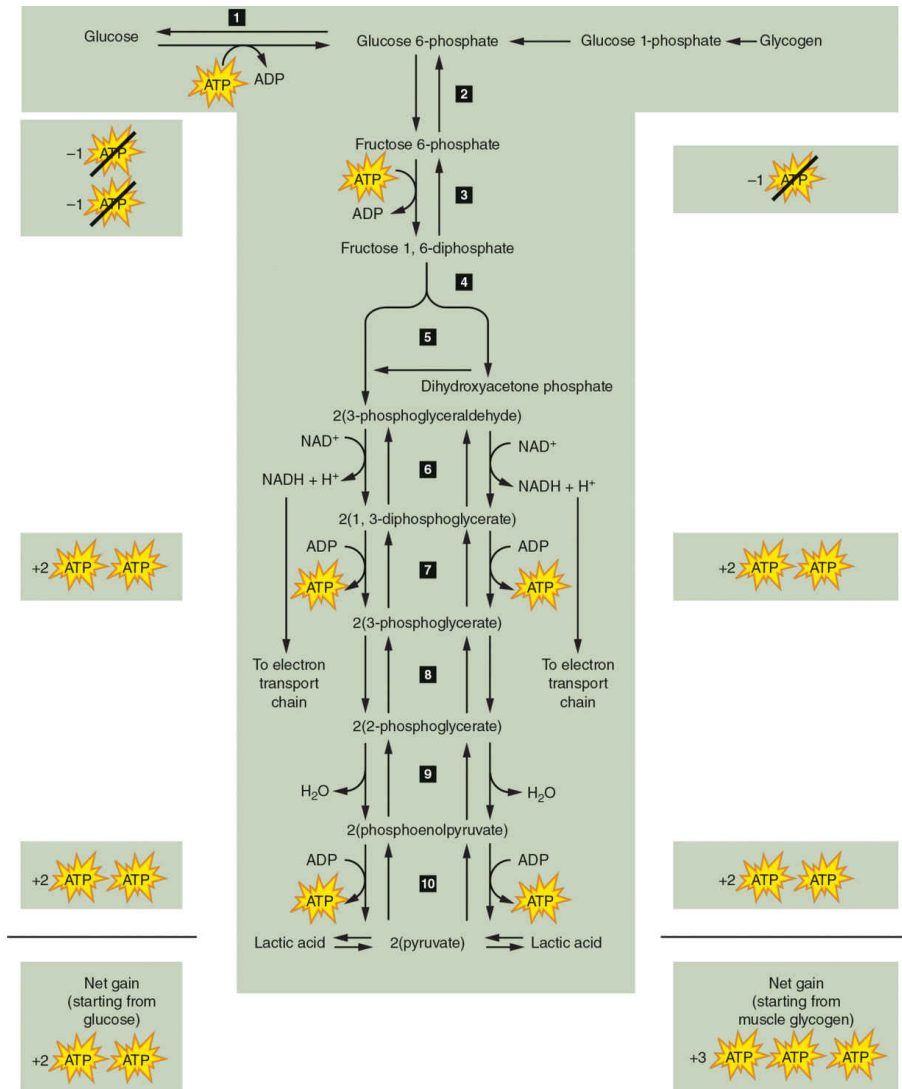


FIGURE 2.7 The derivation of energy (ATP) by glycolysis. Glycolysis involves the breakdown of one glucose (six-carbon) molecule to two three-carbon molecules of pyruvic acid. The process can begin with either glucose circulating in the blood or glycogen (a chain of glucose molecules, the storage form of glucose in muscle and liver). Note that there are roughly 10 separate steps in this anaerobic process, and the net result is the generation of either two or three ATP molecules, depending on whether glucose or glycogen is the initial substrate.



Glucose accounts for about 99% of all sugars circulating in the blood. Blood glucose comes from the digestion of carbohydrate and the breakdown of liver glycogen. Glycogen is synthesized from

glucose by a process called glycogenesis and is stored in the liver or in muscle until needed. At that time, the glycogen is broken down to glucose-1-phosphate, which enters the glycolysis pathway, a process termed **glycogenolysis**.

Before either glucose or glycogen can be used to generate energy, it must be converted to a compound called glucose-6-phosphate. Even though the goal of glycolysis is to release ATP, the conversion of a molecule of glucose to glucose-6-phosphate requires the expenditure or input of one ATP molecule. In the conversion of glycogen, glucose-6-phosphate is formed from glucose-1-phosphate without this energy expenditure. Glycolysis technically begins once the glucose-6-phosphate is formed.

Glycolysis requires 10 to 12 enzymatic reactions for the breakdown of glycogen to pyruvic acid, which is then converted to lactic acid. All steps in the pathway and all of the enzymes involved operate within the cell cytoplasm. The net gain from this process is 3 moles (mol) of ATP formed for each mole of glycogen broken down. If glucose is used instead of glycogen, the gain is only 2 mol of ATP because 1 mol was used for the conversion of glucose to glucose-6-phosphate.

This energy system obviously does not produce large amounts of ATP. Despite this limitation, the combined actions of the ATP-PCr and glycolytic systems allow the muscles to generate force even when the oxygen supply is limited. These two systems predominate during the early minutes of high-intensity exercise.

Another major limitation of anaerobic glycolysis is that it causes an accumulation of lactic acid in the muscles and body fluids. Glycolysis produces pyruvic acid. This process does not require oxygen, but the presence of oxygen determines the fate of the pyruvic acid. Without oxygen present, the pyruvic acid is converted directly to lactic acid, an acid with the chemical formula $C_3H_6O_3$ that quickly dissociates, forming lactate. The terms *pyruvic acid* and *pyruvate*, and *lactic acid* and *lactate*, are often used interchangeably in exercise physiology. In each case, the acid form of the molecule is relatively unstable at normal body pH and rapidly loses a hydrogen ion. The remaining molecule is more correctly called pyruvate or lactate. Lactate can itself be a source of energy as discussed later in this chapter.

In all-out sprint events lasting 1 or 2 min, the demands on the glycolytic system are high, and muscle lactic acid concentrations can increase from a resting value of about 1 mmol/kg of muscle to more than 25 mmol/kg. This acidification of muscle fibers inhibits further glycogen breakdown because it impairs glycolytic enzyme function. In addition, the acid decreases the fibers' calcium-binding capacity and thus may impede muscle contraction.

The rate-limiting enzyme in the glycolytic pathway is **phosphofructokinase (PFK)**. Like almost all rate-limiting enzymes, PFK catalyzes an early step in the pathway, the conversion of fructose-6-phosphate to fructose-1,6-diphosphate. Increasing ADP and P_i concentrations enhance PFK activity and therefore speed up glycolysis, while elevated ATP concentrations slow glycolysis by inhibiting PFK. Additionally, because the glycolytic pathway feeds into the Krebs cycle for additional energy production when oxygen is present (discussed later), products of the Krebs cycle, especially citrate and hydrogen ions, likewise feedback to inhibit PFK.

A muscle fiber's rate of energy use during exercise can be 200 times greater than at rest. The ATP-PCr and glycolytic systems alone cannot supply all the needed energy. Furthermore, these two systems are not capable of supplying all of the energy needs for all-out activity lasting more than 2 min or so. Prolonged exercise relies on the third energy system, the oxidative system.

In Review

- The formation of ATP provides cells with a high-energy compound for storing and, when broken down, releasing energy. It serves as the immediate source of energy for most body functions, including muscle contraction.
- Adenosine triphosphate is generated through three primary energy systems:
 1. The ATP-PCr system
 2. The glycolytic system
 3. The oxidative system
- In the ATP-PCr system, P_i is separated from PCr through the action of creatine kinase. The P_i can then combine with ADP to form ATP using the energy released from the breakdown of PCr. This system is anaerobic, and its main function is to maintain ATP levels early in exercise. The energy yield is 1 mol of ATP per 1 mol of PCr.

- The glycolytic system involves the process of glycolysis, through which glucose or glycogen is broken down to pyruvic acid. When glycolysis occurs without oxygen, the pyruvic acid is converted to lactic acid. One mole of glucose yields 2 mol ATP, but 1 mol glycogen yields 3 mol ATP.
 - The ATP-PCr and glycolytic systems are major contributors of energy during short-burst activities lasting up to 2 min and during the early minutes of longer high-intensity exercise.
-

Oxidative System

The final system of cellular energy production is the **oxidative system**. This is the most complex of the three energy systems, and only a brief overview is provided here. The process by which the body breaks down substrates with the aid of oxygen to generate energy is called cellular respiration. Because oxygen is required, this is an aerobic process. Unlike the anaerobic production of ATP that occurs in the cytoplasm of the cell, the oxidative production of ATP occurs within special cell organelles called **mitochondria**. In muscles, these are adjacent to the myofibrils and are also scattered throughout the sarcoplasm (see [figure 1.3](#)).

The total number, or density, of mitochondria within a muscle fiber is determined by its demand for ATP production, but the precise location of these mitochondria within the cell is determined by oxygen diffusion. Each individual muscle fiber has an optimal distribution of mitochondria within the cell that allows for a near maximal rate of ATP production while exposing the mitochondria to as little excess oxygen as possible. Excess oxygen exposure in mitochondria creates reactive oxygen species (ROS), which are toxic to the cell at high concentrations.^{5,7}

Within a muscle cell, mitochondria tend to be localized along the periphery of the fiber, with higher densities near capillaries. This arrangement functions to create gradients in the oxygen concentration from the capillary to the mitochondria to facilitate the flow of oxygen into the mitochondria. Having mitochondria localized toward the periphery of the cell benefits the muscle fiber by optimizing oxygen delivery to sustain high metabolic rates.⁶ However, having the mitochondria located around the periphery of the cell also increases ROS production because of their exposure to oxygen.

Thus, mitochondria tend to be distributed nonuniformly around the outside of the cell, depending on capillary location, rather than being evenly spaced. This location is optimal for maintaining high metabolic rates while minimizing risk for increasing ROS production, which can negatively affect the cell.

Muscles need a steady supply of energy to continuously produce the force needed during long-term activity. Unlike what happens with anaerobic ATP production, the oxidative system is slow to turn on, but it has a much larger energy-producing capacity, so aerobic metabolism is the primary method of energy production during endurance activities. This places considerable demands on the cardiovascular and respiratory systems to deliver oxygen to the active muscles. Oxidative energy production can come from carbohydrates (starting with glycolysis) or fats.

Oxidation of Carbohydrate

As shown in [figure 2.8](#), oxidative production of ATP from carbohydrates involves three processes:

- Glycolysis ([figure 2.8a](#))
- The Krebs cycle ([figure 2.8b](#))
- The electron transport chain ([figure 2.8c](#))

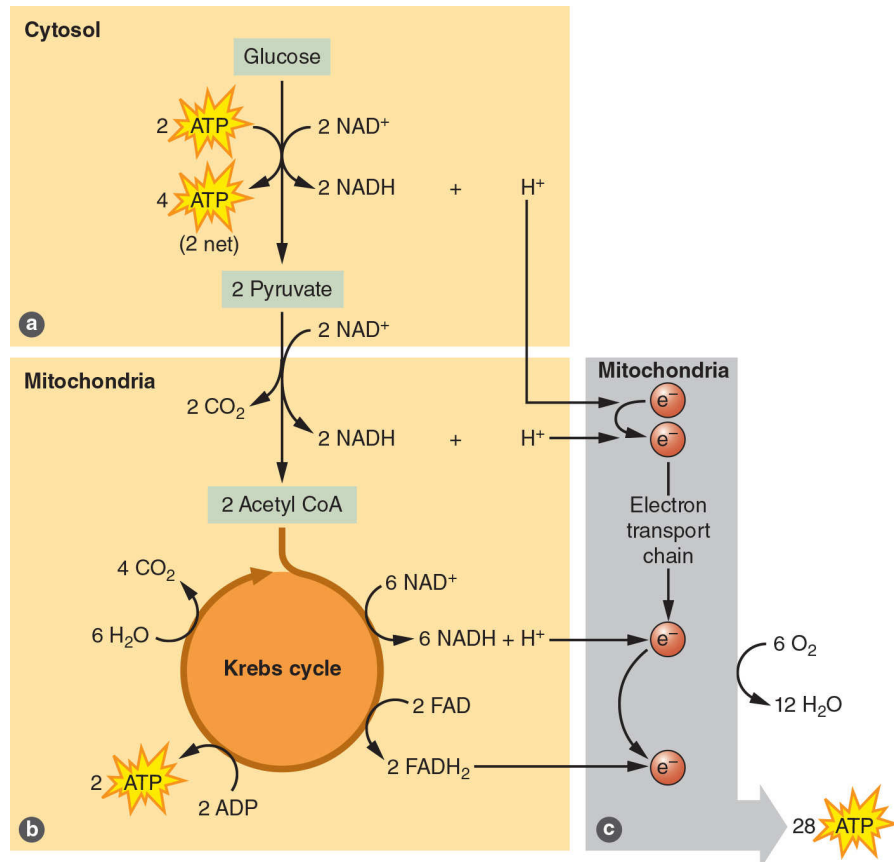


FIGURE 2.8 In the presence of oxygen, after glucose (or glycogen) has been reduced to pyruvate, (a) the pyruvate is first catalyzed to acetyl coenzyme A (acetyl CoA), which can enter (b) the Krebs cycle, where oxidative phosphorylation occurs. Hydrogen ions released during the Krebs cycle then combine with coenzymes that carry the hydrogen ions to (c) the electron transport chain.



Glycolysis In carbohydrate metabolism, glycolysis plays a role in *both* anaerobic and aerobic ATP production. The process of glycolysis is the same regardless of whether oxygen is present. The presence of oxygen determines only the fate of the end product, pyruvic acid. Recall that anaerobic glycolysis produces lactic acid and only three net moles of ATP per mole of glycogen, or two net moles of ATP per mole of glucose. In the presence of oxygen, however, the pyruvic acid is converted into a compound called **acetyl coenzyme A (acetyl CoA)**.

Krebs Cycle Once formed, acetyl CoA enters the **Krebs cycle** (also called the citric acid cycle or tricyclic acid cycle), a complex series of chemical reactions that permit the complete oxidation of acetyl CoA (see [figure 2.9](#)). Recall that for every glucose molecule that enters the glycolytic pathway, two molecules of pyruvate are formed. Therefore, each glucose molecule that begins the energy-producing process in the presence of oxygen results in two complete Krebs cycles.

As depicted in 2.8*b* (and shown in more detail in [figure 2.9](#)), the conversion of succinyl CoA to succinate in the Krebs cycle results in the generation of guanosine triphosphate, or GTP, a high-energy compound similar to ATP. Guanosine triphosphate then transfers a P_i to ADP to form ATP. These two ATPs (per molecule of glucose) are formed by substrate-level phosphorylation. So at the end of the Krebs cycle, two additional moles of ATP have been formed directly, and the original carbohydrate has been broken down into carbon dioxide and hydrogen.

As in the other pathways involved in energy metabolism, Krebs cycle enzymes are regulated by negative feedback at several steps in the cycle. The rate-limiting enzyme in the Krebs cycle is isocitrate dehydrogenase, which, like PFK, is inhibited by ATP and activated by ADP and P_i as is the electron transport chain. Because muscle contraction relies on the availability of calcium in the cell, excess calcium also stimulates the rate-limiting enzyme isocitrate dehydrogenase.

Electron Transport Chain During glycolysis, hydrogen ions are released when glucose is metabolized to pyruvic acid. Additional hydrogen ions are released in the conversion of pyruvate to acetyl CoA and at several steps in the Krebs cycle. If these hydrogen ions remained in the system, the inside of the cell would become too acidic. What happens to this hydrogen?

The Krebs cycle is coupled to a series of reactions known as the **electron transport chain** ([figure 2.8c](#)). The hydrogen ions released during the processes of glycolysis, the conversion of pyruvic acid to acetyl CoA, and the Krebs cycle combine with two coenzymes: nicotinamide adenine dinucleotide (NAD) and flavin adenine dinucleotide (FAD), converting each to its reduced form (NADH and $FADH_2$, respectively). During each Krebs cycle, three molecules of

NADH and one molecule of FADH_2 are produced. These carry the hydrogen atoms (electrons) to the electron transport chain, a group of mitochondrial protein complexes located in the inner mitochondrial membrane (figure 2.10).

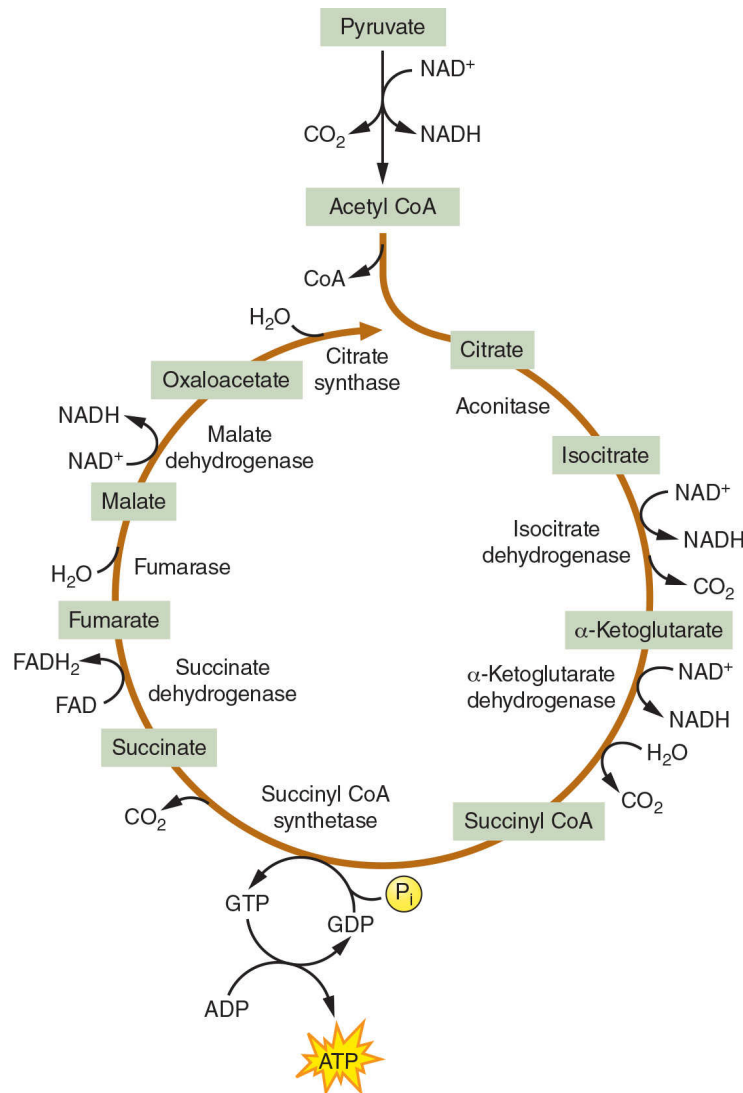


FIGURE 2.9 The series of reactions that take place during the Krebs cycle, showing the compounds formed and enzymes involved.



These protein complexes contain a series of enzymes and iron-containing proteins known as **cytochromes**. These proteins serve as electron magnets that transfer electrons, where the first complex, flavin mononucleotide (FMN), is a stronger magnet for electrons than NADH, the second complex is a stronger magnet than the first, and so on down the chain. As high-energy electrons are passed from complex to complex along this chain, some of the energy released by those reactions is used to pump H^+ from the mitochondrial matrix into the outer mitochondrial compartment. As these hydrogen ions move back across the membrane down their concentration gradient, energy is transferred to ADP, and ATP is formed. This final step requires an enzyme known as ATP synthase. At the end of the chain, the H^+ combines with oxygen to form water, thus preventing acidification of the cell. This is illustrated in [figure 2.11](#). Because this overall process relies on oxygen as the final acceptor of electrons and H^+ , it is referred to as oxidative phosphorylation.

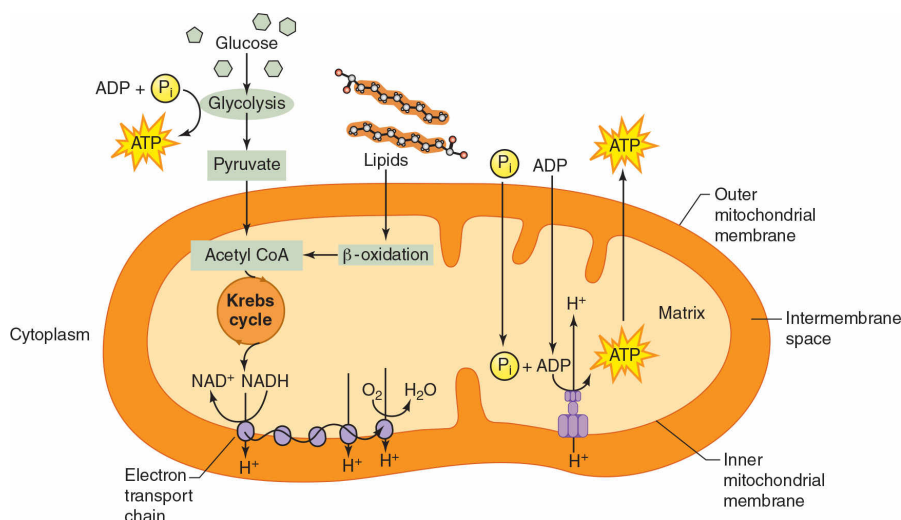


FIGURE 2.10 Locations of the processes of glycolysis (cytoplasm), the Krebs cycle (mitochondria), and the electron transport chain (inner mitochondrial membrane).



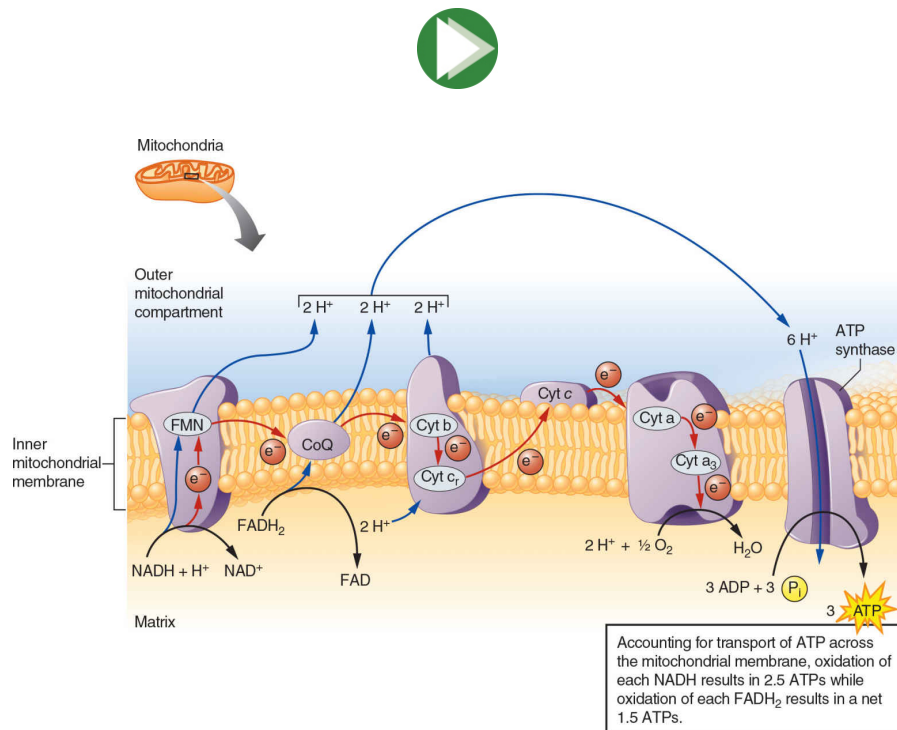


FIGURE 2.11 The final step in the aerobic production of adenosine triphosphate (ATP) is the transfer of energy from the high-energy electrons of reduced nicotinamide adenine dinucleotide (NADH) and reduced flavin adenine dinucleotide (FADH_2) within the mitochondria, following a series of steps known as the electron transport chain.

For every pair of electrons transported to the electron transport chain by NADH, three molecules of ATP are formed, while the electrons passed through the electron transport chain by FADH_2 yield only two molecules of ATP. However, because the NADH and FADH_2 are outside the membrane of the mitochondria, the H^+ must be shuttled through the membrane, which requires energy to be used. So, in reality, the net yields are only 2.5 ATPs per NADH and 1.5 ATPs per FADH_2 .

Energy Yield From Oxidation of Carbohydrate The complete oxidation of glucose can generate 32 molecules of ATP, while 33 ATPs are produced from one molecule of muscle glycogen. The sites of ATP production are summarized in [figure 2.12](#). The net production of ATP

from substrate-level phosphorylation in the glycolytic pathway leading into the Krebs cycle results in a net gain of two ATPs (or three from glycogen). A total of 10 NADH molecules leading into the electron transport chain—two in glycolysis, two in the conversion of pyruvic acid to acetyl CoA, and six in the Krebs cycle—yields 25 net ATP molecules. Remember that while 30 ATPs are actually produced, the energy cost of transporting ATP across membranes uses five of those ATPs. The two FAD molecules in the Krebs cycle that are involved in electron transport result in three additional net ATPs. And finally, substrate-level phosphorylation within the Krebs cycle involving the molecule GTP adds another two ATP molecules.

Accounting for the energy cost of shuttling electrons across the mitochondrial membrane is a relatively new concept in exercise physiology, and many textbooks still refer to net energy productions of 36 to 39 ATPs per molecule of glucose.

Oxidation of Fat

As noted earlier, fat also contributes importantly to the muscle's energy needs. Muscle and liver glycogen stores can provide only ~2,500 kcal of energy, but the fat stored inside muscle fibers and in fat cells can supply at least 70,000 to 75,000 kcal, even in a lean adult. Although many chemical compounds (such as triglycerides, phospholipids, and cholesterol) are classified as fats, only triglycerides are major energy sources. Triglycerides are stored in fat cells and between and within skeletal muscle fibers. To be used for energy, a triglyceride must be broken down to its basic units: one molecule of glycerol and three FFA molecules. This process is called **lipolysis** and is controlled by enzymes known as lipases.

common intermediate through which all substrates enter the Krebs cycle for oxidative metabolism.

β -Oxidation is a series of steps in which two-carbon acyl units are chopped off of the carbon chain of the FFA. The acyl units become acetyl CoA, which then enters the Krebs cycle for the formation of ATP. The number of steps depends on the number of carbons in the FFA, usually between 14 and 24 carbons. For example, if an FFA originally has a 16-carbon chain, β -oxidation yields eight molecules of acetyl CoA.

On entering the muscle fiber, FFAs must be enzymatically activated with energy from ATP, preparing them for catabolism (breakdown) within the mitochondria. Like glycolysis, β -oxidation requires an input energy of two ATPs for activation but, unlike glycolysis, produces no ATPs directly.

Krebs Cycle and the Electron Transport Chain After β -oxidation, fat metabolism follows the same path as oxidative carbohydrate metabolism. Acetyl CoA formed by β -oxidation enters the Krebs cycle. The Krebs cycle generates hydrogen, which is transported to the electron transport chain along with the hydrogen generated during β -oxidation to undergo oxidative phosphorylation. As in glucose metabolism, the by-products of FFA oxidation are ATP, H_2O , and carbon dioxide (CO_2). However, the complete combustion of an FFA molecule requires more oxygen because an FFA molecule contains considerably more carbon molecules than a glucose molecule.

The advantage of having more carbon molecules in FFAs than in glucose is that more acetyl CoA is formed from the metabolism of a given amount of fat, so more acetyl CoA enters the Krebs cycle and more electrons are sent to the electron transport chain. This is why fat metabolism can generate much more energy than glucose metabolism. Unlike glucose or glycogen, fats are heterogeneous, and the amount of ATP produced depends on the specific fat oxidized.

RESEARCH PERSPECTIVE 2.1

White, Brown, and (Perhaps) Beige Fat in Humans

Brown adipose tissue (BAT), often called brown fat, is found in almost every species of mammal, especially in those that hibernate. Unlike white adipose tissue, which is specialized for lipid storage and breakdown (lipolysis) to meet

long-duration metabolic demands, the function of brown adipose is to transfer energy from food directly into heat. Brown adipose cells contain many small lipid droplets and many mitochondria, which give the tissue its brown appearance. BAT cells also have more blood vessels than white adipose cells to supply the tissue with oxygen and nutrients and distribute the heat produced in the cells to the rest of the body. The inner membrane of the mitochondria of BAT cells has a specialized protein called uncoupling protein that uncouples the electron transport chain from the creation of ATP (phosphorylation). While white fat generates ATP for energy, brown fat's primary role is to produce heat and increase metabolism, especially at rest.

Brown adipose is abundant in newborn babies and young children. However, for a long time, it was believed that brown adipose stores were absent in adult humans. In 2009, a study published in the *New England Journal of Medicine* showed that adult humans have functionally active brown adipose tissue.³ Using positron-emission tomography and computed tomography (PET-CT) scans, researchers found that the most common location for this brown adipose tissue in adults was near the clavicles, that brown adipose was more frequently found in women than in men, and that individuals with a higher body mass index have less brown fat. Because BAT promotes energy dissipation rather than energy storage (the role of white adipose tissue), its discovery in humans sparked great interest in the possibility of increasing the activity of BAT to target diseases like obesity and type 2 diabetes.

Several studies have recently reported that chronic endurance exercise may promote the expression of similar thermogenic (heat-producing) genes in white adipose tissue, resulting in the browning of white fat. In one animal study, training-induced changes in fat type resulted in increases in resting energy expenditure of up to 17% in trained rats.² It is still unclear whether exercise training increases BAT mass or promotes the browning of white adipose tissue in humans, but studies are now underway to use the metabolic potential of BAT to increase whole-body energy expenditure. The ultimate goal of these studies is to treat obesity and other metabolic diseases.

Consider the example of palmitic acid, a rather abundant 16-carbon FFA. The combined reactions of oxidation, the Krebs cycle, and the electron transport chain produce 106 molecules of ATP from one molecule of palmitic acid (see [table 2.2](#)), compared with only 32 molecules of ATP from glucose or 33 from glycogen.

Oxidation of Protein

As noted earlier, carbohydrates and fatty acids are the preferred fuel substrates. But proteins, or rather the amino acids that compose

proteins, are also used for energy under some circumstances. Some amino acids can be converted into glucose, a process called gluconeogenesis (see [figure 2.1](#)). Alternatively, some can be converted into various intermediates of oxidative metabolism (such as pyruvate or acetyl CoA) to enter the oxidative process.

Protein's energy yield is not as easily determined as that of carbohydrate or fat because protein also contains nitrogen. When amino acids are catabolized, some of the released nitrogen is used to form new amino acids, but the remaining nitrogen cannot be oxidized by the body. Instead it is converted into urea and then excreted, primarily in the urine. This conversion requires the use of ATP, so some energy is spent in this process.

When protein is broken down through combustion in the laboratory, the energy yield is 5.65 kcal/g. However, because of the energy expended in converting nitrogen to urea when protein is metabolized in the body, the energy yield is only about 4.1 kcal/g.

To accurately assess the rate of protein metabolism, the amount of nitrogen being eliminated from the body must be determined. This requires urine collection for 12 to 24 h periods, a time-consuming process. Because the healthy body uses little protein during rest and exercise (usually not more than 10% of total energy expended), estimates of total energy expenditure generally ignore protein metabolism.

TABLE 2.2 ATP Produced From One Molecule of Palmitic Acid

Stage of process	Direct (substrate-level oxidation)	By oxidative phosphorylation
Fatty acid activation	0	-2
β -oxidation (occurs 7 times)	0	28
Krebs cycle (occurs 8 times)	8	72
Subtotal	8	98
Total	106	

Lactic Acid as a Source of Energy During Exercise

Lactic acid is in a state of constant turnover within cells, being produced by glycolysis and removed from the cell, primarily through oxidation. Thus, despite its reputation as a cause of fatigue, lactic acid can be, and is, used as an actual fuel source during exercise. This occurs through several mechanisms.

First, we now know that lactate produced by glycolysis in the cytoplasm of a muscle fiber can be taken up by the mitochondria within that same fiber and directly oxidized. This occurs mostly in cells with a high density of mitochondria like type I (high oxidative) muscle fibers, cardiac muscle, and liver cells.

Second, lactate produced in a muscle fiber can be transported away from its site of production and used elsewhere by a process called the lactate shuttle, first described by Dr. George Brooks. Lactate is produced primarily by type II muscle fibers but can be transported to adjacent type I fibers by diffusion or active transport. In that regard, most of the lactate produced in a muscle never leaves that muscle. It can also be transported through the circulation to sites where it can be directly oxidized. The lactate shuttle allows for glycolysis in one cell to supply fuel for use by another cell. Special transporters called monocarboxylate transport (MCT) proteins facilitate the movement of lactate between cells and tissues and likely within cells. During exercise, approximately 80% to 90% of lactate is transferred across the sarcolemma either by passive diffusion or by facilitated transport through MCTs. These transporters can be expressed in differing numbers, depending on the properties of the cells and tissues helping to move lactate in the cells that are the most metabolically active. Using lactate as a metabolic fuel accounts for approximately 70% to 75% of lactate removal during exercise.

Finally, some of the lactic acid produced in the muscle is transported by the blood to the liver, where it is reconverted to pyruvic acid and back to glucose (gluconeogenesis) and transported back to the working muscle. This is called the Cori cycle. Without this recycling of lactate into glucose for use as an energy source, prolonged exercise would be severely limited. On a more integrative level, lactate produced in exercising skeletal muscle is taken up and oxidized in the brain. Thus, lactate not only is integrally involved as a metabolic fuel but also responds to changes in nutrient sensing as different metabolic fuels are used during exercise.

Summary of Substrate Metabolism

As shown in [figure 2.13](#), the ability to produce muscle contraction for exercise is a matter of energy supply and energy demand. Both the contraction of skeletal muscle fibers and their relaxation require

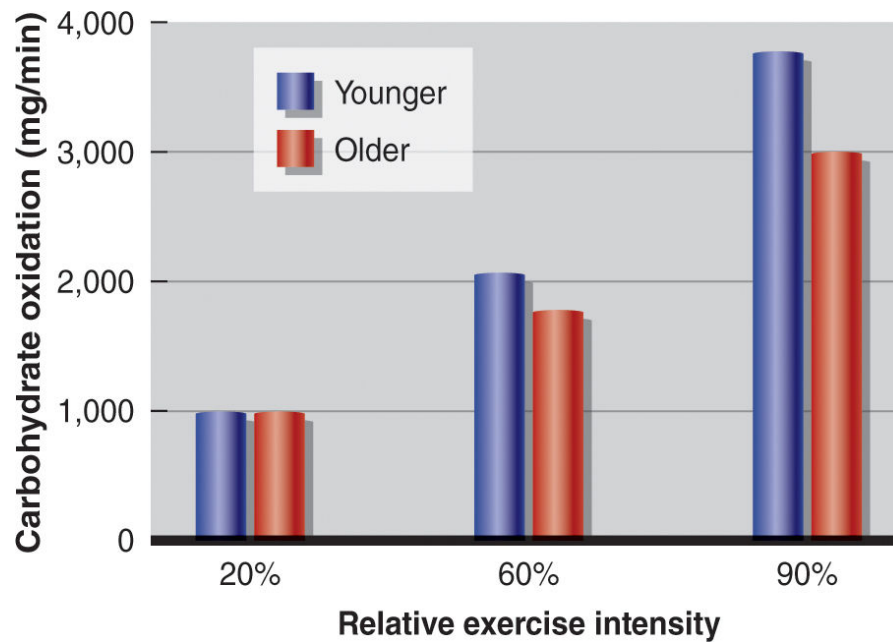
energy. That energy comes from foodstuffs in the diet and stored energy in the body. The ATP-PCr system operates within the cytosol of the cell, as does glycolysis, and neither requires oxygen for ATP production. Oxidative phosphorylation takes place within the mitochondria. Note that under aerobic conditions, both major substrates—carbohydrates and fats—are reduced to the common intermediate acetyl CoA that enters the Krebs cycle.

RESEARCH PERSPECTIVE 2.2

Lifelong Training Can Lead to More Efficient Fuel Utilization

While aging is associated with a decrease in exercise capacity and sport performance, it is clear that the skeletal muscle of older adults can be trained to meet the demands of exercise. Remaining physically active throughout a person's life protects against some of the age-related decrements in muscle-fiber size, fiber type, mitochondrial number, and oxidative capacity when compared to older sedentary people. These age-related changes and adaptations are discussed in greater detail in [chapter 18](#).

A recent study performed at the University of Pittsburgh sought to determine whether the skeletal muscles of older masters athletes had the same substrate storage and capacity for oxidation of those fuels as those of younger athletes who trained similarly (i.e., used the same mode of exercise and frequency of training).⁴ That study found that lifelong masters athletes have greater triglyceride stores in their muscle fibers and a greater proportion of oxidative fibers compared to the young athletes. These differences resulted in better metabolic efficiency—a lower reliance on carbohydrate oxidation—during exercise at high intensities in the older athletes (see figure). Lifelong endurance exercise protects against some of the age-associated decreases in oxidative potential and provides older athletes with an increased capacity for fat oxidation to produce ATP during exercise.



A simplified illustration of carbohydrate oxidation at different relative exercise intensities in younger and older adult athletes. Lifelong masters endurance athletes rely less on carbohydrate oxidation at higher exercise intensities compared to similarly trained younger athletes.

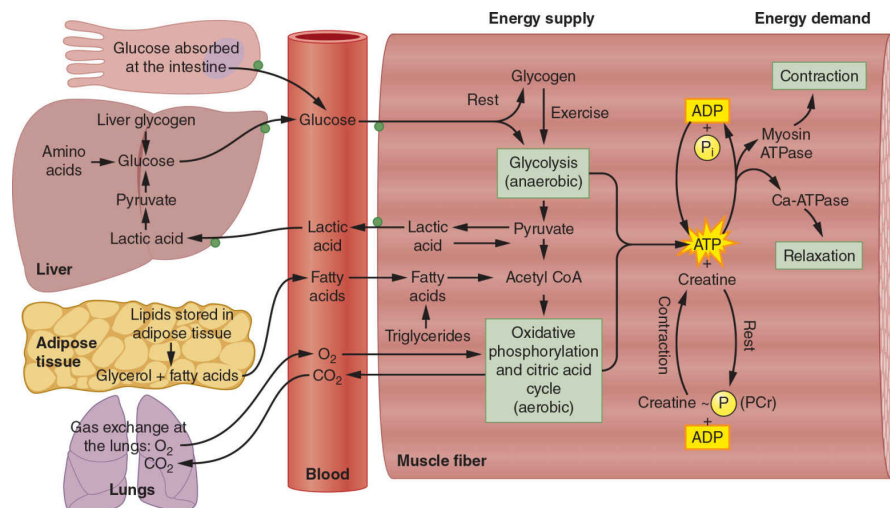


FIGURE 2.13 The metabolism of carbohydrate, fat, and (to a lesser extent) protein shares some common pathways within the muscle fiber. The adenosine triphosphate (ATP) molecules generated by oxidative and nonoxidative metabolism are used by those steps in muscle contraction and relaxation that demand energy.

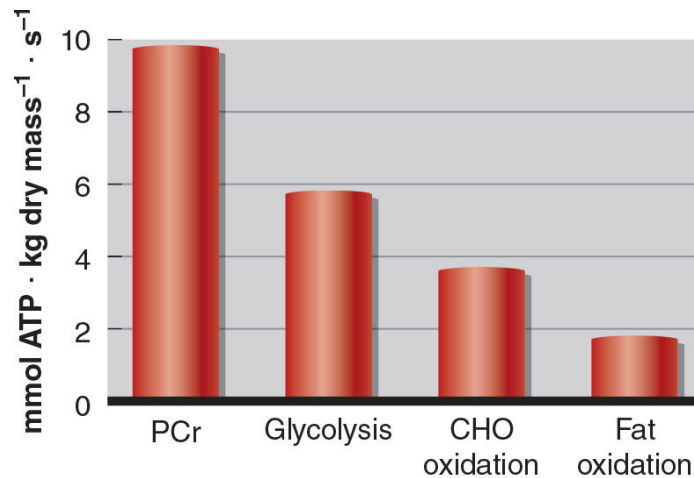


In Review

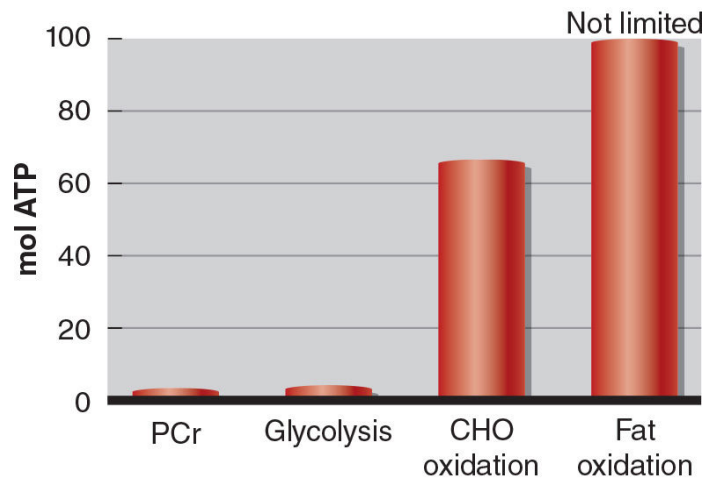
- The oxidative system involves the breakdown of substrates in the presence of oxygen. This system yields more energy than the ATP-PCr or the glycolytic system.
- Oxidation of carbohydrate involves glycolysis, the Krebs cycle, and the electron transport chain. The end result is H_2O , CO_2 , and 32 or 33 ATP molecules per carbohydrate molecule.
- Fat oxidation begins with β -oxidation of FFAs and then follows the same path as carbohydrate oxidation: acetyl CoA moving into the Krebs cycle and the electron transport chain. The energy yield for fat oxidation is much higher than for carbohydrate oxidation, and it varies with the FFA being oxidized. However, the maximum rate of high-energy phosphate formation from lipid oxidation is too low to match the rate of utilization of high-energy phosphate during higher-intensity exercise, and the energy yield of fat per oxygen molecule used is much less than that for carbohydrate.
- Although fat provides more kilocalories of energy per gram than carbohydrate, fat oxidation requires more oxygen than carbohydrate oxidation. The energy yield from fat is 5.6 ATP molecules per oxygen molecule used, compared with carbohydrate's yield of 6.3 ATP per oxygen molecule. Oxygen delivery is limited by the oxygen transport system, so carbohydrate is the preferred fuel during high-intensity exercise.
- The maximum rate of ATP production from lipid oxidation is too low to match the rate of utilization of ATP during high-intensity exercise. This explains the reduction in an athlete's race pace when carbohydrate stores are depleted and fat, by default, becomes the predominant fuel source.
- Measurement of protein oxidation is more complex because amino acids contain nitrogen, which cannot be oxidized. Protein contributes relatively little to energy production, generally less than 10%, so its metabolism is often considered negligible.
- Despite its reputation as a potential factor in causing fatigue, lactic acid can be, and is, used as an important fuel source during exercise.

Interaction of the Energy Systems

The three energy systems do not work independently of one another, and no activity is 100% supported by any single energy system. When a person exercises at the highest intensity possible, from the shortest sprints (less than 10 s) to endurance events (greater than 30 min), each of the energy systems is contributing to the total energy needs of the body. Generally, one energy system dominates energy production, except when there is a transition from the predominance of one energy system to another. As an example, in a 10 s, 100 m sprint, the ATP-PCr system is the predominant energy system, but both the anaerobic glycolytic and the oxidative systems provide a small portion of the energy needed. At the other extreme, in a 30 min, 10,000 m (10,936 yd) run, the oxidative system is predominant, but both the ATP-PCr and anaerobic glycolytic systems contribute some energy as well.



a Maximal rate of ATP generation



b Maximal available energy

FIGURE 2.14 The various energy systems have a reciprocal relation with respect to (a) the maximal rate at which energy can be produced and (b) the capacity to produce that energy.



Figure 2.14 shows the reciprocal relation among the energy systems with respect to power and capacity. The ATP-PCr energy system can provide energy at a fast rate but has a very low capacity for energy production. Thus it supports exercise that is intense but of very short duration. By contrast, fat oxidation takes longer to gear up

and produces energy at a slower rate; however, the amount of energy it can produce is unlimited. The characteristics of the muscle fiber's energy systems are listed in [table 2.3](#).

The Crossover Concept

The **crossover concept** was first described by Brooks and Mercier¹ to describe the relative balance between carbohydrate (CHO) and fat metabolism during sustained exercise. At rest and during exercise at low to moderate intensities (below 60% of maximal oxygen uptake), lipids serve as the main substrate for generating ATP. During high-intensity exercise (above 75% of maximal oxygen uptake), increases in muscle glycogenolysis and the recruitment of more type II muscle fibers promote a shift to CHO as the predominant substrate for generating ATP. The crossover point is the intensity where fat and carbohydrate utilization intersect ([figure 2.15](#)) as the energy from fat decreases and the energy from carbohydrate increases. Beyond this crossover point, further increases in power are met with further increments in CHO utilization and decrements in fat oxidation.

The crossover point is affected by both the exercise intensity and endurance training status. Endurance training results in biochemical adaptations within the muscle fibers that promote and support oxidation of FFAs, including an increase in the number of mitochondria, increased oxidative enzymes, and changes in β -oxidation and the electron transport chain—all important determinants of fat metabolism. The result of training is to allow the body to spare muscle glycogen, since carbohydrate stores within the body are limited. These training-induced adaptations shift the crossover point toward higher exercise intensities. Diet (energy supply and stores) and prior exercise play secondary roles in determining the balance of substrates utilization during submaximal exercise.

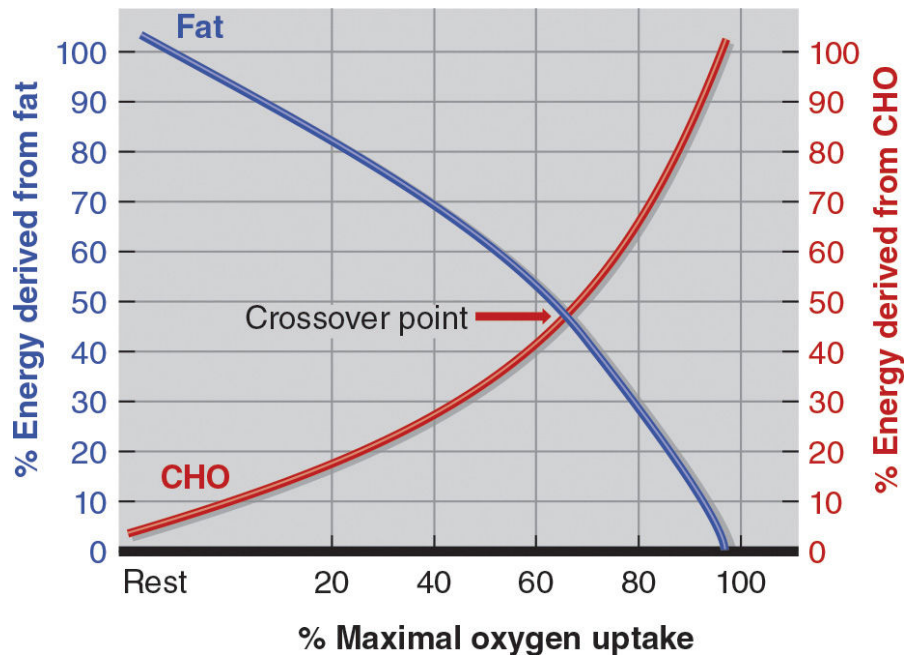


FIGURE 2.15 The relation between the relative contributions of fat and carbohydrate (CHO) utilization to overall energy expenditure as a function of exercise intensity. The point at which the two lines intersect illustrates the classic crossover concept.



TABLE 2.3 Characteristics of the Various Energy Supply Systems

Energy system	Oxygen necessary?	Overall chemical reaction	Relative rate of ATP formed per second	ATP formed per molecule of substrate	Available capacity
ATP-PCr	No	PCr to Cr	10	1	<15 s
Glycolysis	No	Glucose or glycogen to lactate	5	2-3	~1 min
Oxidative (from carbohydrate)	Yes	Glucose or glycogen to CO ₂ and H ₂ O	2.5	36-39*	~90 min
Oxidative (from fat)	Yes	FFA or triglycerides to CO ₂ and H ₂ O	1.5	>100	Days

*Production of 36 to 39 ATP per molecule of carbohydrate excludes energy cost of transport through membranes. The net production is slightly lower (see text).

Courtesy of Dr. Martin Gibala, McMaster University, Hamilton, Ontario, Canada.

The Oxidative Capacity of Muscle

We have seen that the processes of oxidative metabolism have the highest energy yields. It would be ideal if these processes always

functioned at peak capacity. But, as with all physiological systems, they operate within certain constraints. The oxidative capacity of muscle ($\dot{Q} O_2$) is a measure of its maximal capacity to use oxygen. This measurement is made in the laboratory, where a small amount of muscle tissue can be tested to determine its capacity to consume oxygen when chemically stimulated to generate ATP. A muscle's oxidative capacity ultimately depends on its oxidative enzyme concentrations, fiber type composition, and oxygen availability.

Enzyme Activity

The capacity of muscle fibers to oxidize carbohydrate and fat is difficult to determine. Numerous studies have shown a close relation between a muscle's ability to perform prolonged aerobic exercise and the activity of its oxidative enzymes. Because many different enzymes are required for oxidation, the enzyme activity of the muscle fibers provides a reasonable indication of their oxidative potential.

Measuring all the enzymes in muscles is impossible, so a few representative enzymes have been selected to reflect the aerobic capacity of the fibers. The enzymes most frequently measured are succinate dehydrogenase and citrate synthase, mitochondrial enzymes involved in the Krebs cycle (see [figure 2.9](#)). [Figure 2.16](#) illustrates the close correlation between succinate dehydrogenase activity in the vastus lateralis muscle and that muscle's oxidative capacity. Endurance athletes' muscles have oxidative enzyme activities two to four times greater than those of untrained men and women.

Fiber Type Composition and Endurance Training

A muscle's fiber type composition primarily determines its oxidative capacity. As noted in [chapter 1](#), type I (slow-twitch) fibers have a greater capacity for aerobic activity than type II (fast-twitch) fibers because type I fibers have more mitochondria and higher concentrations of oxidative enzymes. Type II fibers are better suited for glycolytic energy production. Thus, in general, the more type I fibers in one's muscles, the greater the oxidative capacity of those muscles. Elite distance runners, for example, possess more type I fibers, more mitochondria, and higher muscle oxidative enzyme activities than do untrained individuals.

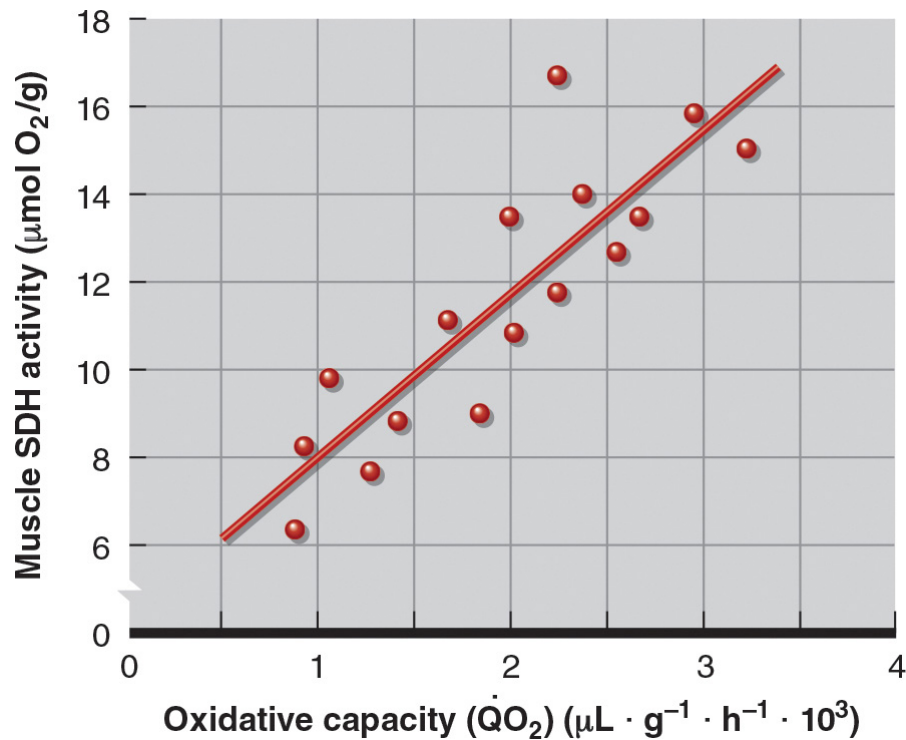


FIGURE 2.16 The relation between muscle succinate dehydrogenase (SDH) activity and its oxidative capacity ($\dot{Q}O_2$), measured in a muscle biopsy sample taken from the vastus lateralis.



Endurance training enhances the oxidative capacity of all fibers, especially type II fibers. Training that places demands on oxidative phosphorylation stimulates the muscle fibers to develop more mitochondria, larger mitochondria, and more oxidative enzymes per mitochondrion. By increasing the fibers' enzymes for β -oxidation, this training also enables the muscle to rely more on fat for aerobic ATP production. Thus, with endurance training, even people with large percentages of type II fibers can increase their muscles' aerobic capacities. But it is generally agreed that an endurance-trained type II fiber will not develop the same high endurance capacity as a similarly trained type I fiber.



Oxygen Needs

Although the oxidative capacity of a muscle is determined by the number of mitochondria and the amount of oxidative enzymes present, oxidative metabolism ultimately depends on an adequate supply of oxygen. At rest, the need for ATP is relatively small, requiring minimal oxygen delivery. As exercise intensity increases, so do energy demands. To meet them, the rate of oxidative ATP production increases. In an effort to meet the muscles' need for oxygen, the rate and depth of respiration increase, improving gas exchange in the lungs, and the heart beats faster and more forcefully, pumping more oxygenated blood to the muscles. Arterioles dilate to facilitate delivery of arterial blood to muscle capillaries.

The human body stores little oxygen. Therefore, the amount of oxygen entering the blood as it passes through the lungs is directly proportional to the amount used by the tissues for oxidative metabolism. Consequently, a reasonably accurate estimate of aerobic energy production can be made by measuring the amount of oxygen consumed at the lungs (see [chapter 5](#)).

RESEARCH PERSPECTIVE 2.3

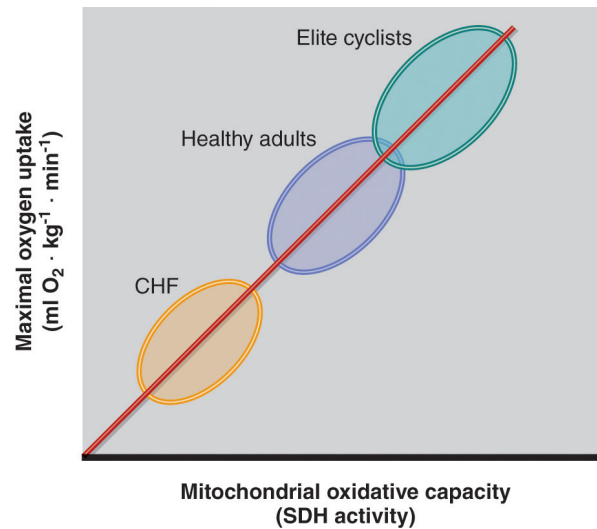
Does the Muscle Fiber's Oxidative Capacity Determine Fitness Level?

Maximal oxygen uptake ($\dot{V}O_{2\max}$; discussed in detail in [chapter 5](#)) is a measurement of cardiorespiratory fitness, so it is not surprising that well-trained endurance athletes have a high $\dot{V}O_{2\max}$. The ability to take in and use oxygen during aerobic exercise may be limited by any number of factors along the pathway of the O_2 molecule as it moves from the atmosphere to the mitochondria to be used for energy: pulmonary ventilation, the oxygen-carrying capacity of the blood, and blood flow to exercising muscle, to name a few. Maximal oxygen uptake is also an important predictor of health, and reductions in $\dot{V}O_{2\max}$ are associated with a loss of mobility and independence in the elderly and an increase in mortality in many chronic diseases. Because of its critical role, exercise physiologists are keenly interested in the factors that limit $\dot{V}O_{2\max}$ in all people, from chronic heart failure patients to professional endurance athletes.

Since the early development of measurement techniques to quantify $\dot{V}O_{2\max}$ in humans, researchers have designed studies to systematically examine each point along the oxygen delivery pathway from inspired air to the mitochondria within muscle fibers. Because it is well accepted that increasing oxygen *supply* to the working muscle improves $\dot{V}O_{2\max}$ and exercise capacity, many scientists believed that the ability of the mitochondria themselves to *use* oxygen—mitochondrial oxidative capacity—was not a limiting factor for maximal oxygen uptake. However, a recent study examined how well the mitochondrial oxidative capacity alone was associated with $\dot{V}O_{2\max}$ across people of vastly different fitness levels.⁸

In that study, researchers measured $\dot{V}O_{2\max}$ during cycling exercise in chronic heart failure patients, healthy subjects, and elite cyclists. They then took muscle biopsy samples from the quadriceps of each subject to measure mitochondrial oxidative capacity. To quantify the muscle fibers' capacity to utilize oxygen, they measured the activity of an important enzyme in the Krebs cycle, succinate dehydrogenase. Interestingly, they found that this measure of mitochondrial oxidative capacity was related to $\dot{V}O_{2\max}$ across all subjects,

regardless of fitness or health status (see figure). Their results indicated that while limitations in oxygen supply certainly limit $\dot{V}O_{2\max}$, maximal oxygen uptake during whole-body exercise is partially determined at the level of the muscle fiber itself.



Simplified figure showing the relation between mitochondrial oxidative capacity, measured as succinate dehydrogenase activity in skeletal muscle biopsy samples obtained from the quadriceps after cycle exercise, and maximal oxygen uptake in chronic heart failure patients (CHF), healthy adults, and elite cyclists. Maximal oxygen uptake is closely related to mitochondrial oxidative capacity across all three subject groups.

IN CLOSING

In this chapter, we focused on energy metabolism and the synthesis of the storage form of energy in the body, ATP. We described in some detail the three basic energy systems used to generate ATP and their regulation and interaction. Finally, we highlighted the important role that oxygen plays in the sustained generation of ATP for continued muscle contraction and the three fiber types found in human skeletal muscle. We next look at the neural control of exercising muscle.

KEY TERMS

acetyl coenzyme A (acetyl CoA)

activation energy

adenosine diphosphate (ADP)

aerobic metabolism

anaerobic metabolism

ATP-PCr system

β -oxidation

bioenergetics

carbohydrate

catabolism

creatine kinase

crossover concept

cytochrome

electron transport chain

enzyme

free fatty acids (FFAs)

gluconeogenesis

glucose

glycogen

glycogenolysis

glycolysis

kilocalories (kcal)

Krebs cycle

lipogenesis

lipolysis

metabolism

mitochondria

negative feedback

oxidative phosphorylation

oxidative system
phosphocreatine (PCr)
phosphofructokinase (PFK)
phosphorylation
rate-limiting enzyme
substrate
triglycerides

STUDY QUESTIONS

1. What is ATP, how is it formed, and how does it provide energy during metabolism?
2. What is the primary substrate used to provide energy at rest? During high-intensity exercise?
3. What is the role of PCr in energy production, and what are its limitations? Describe the relationship between muscle ATP and PCr during sprint exercise.
4. Describe the essential characteristics of the three energy systems.
5. Why are the ATP-PCr and glycolytic energy systems considered anaerobic?
6. What role does oxygen play in the process of aerobic metabolism?
7. Describe the by-products of energy production from ATP-PCr, glycolysis, and oxidation.
8. What is lactic acid, and why is it important?
9. Discuss the interaction among the three energy systems with respect to the rate at which energy can be produced and the sustained capacity to produce that energy.
10. What is meant by the crossover concept, and how does it change with endurance exercise training?
11. How do type I muscle fibers differ from type II fibers in their respective oxidative capacities? What accounts for those differences?

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

www.HumanKinetics.com/PhysiologyOfSportAndExercise

The **KEY TERMS** activity reviews important terms, and the end-of-chapter **QUIZ** tests your understanding of the material covered in the chapter.

